

Fabien Depoilly

Personal Information

Name: Depoilly
First name: Fabien
Date of birth: March 23, 1999 (27 years old)
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Professional Experience

Polytech Lyon *December 2025 – August 2026*
Temporary Teaching and Research Assistant (ATER)

Lyon 1 University Claude Bernard *December 2022 – December 2025*
PhD Candidate, with Supplementary Teaching Duties (ACE)

Academic Background

Lyon 1 University Claude Bernard *December 2022 – June 2026*
PhD Thesis

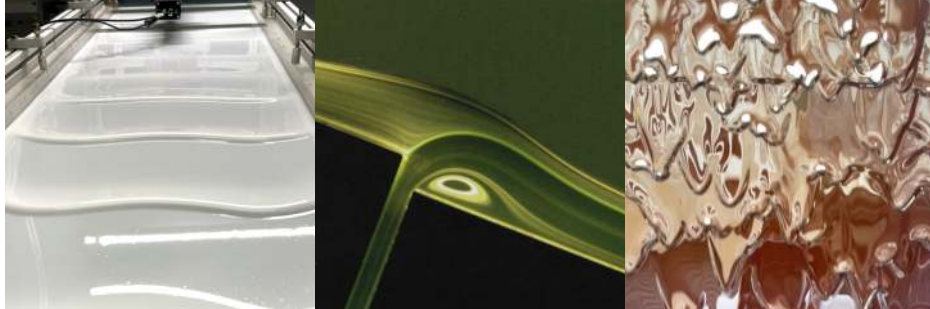
- **Defended on June 30, 2026**
- **Title:** Stability of generalised Newtonian fluid film flows down an inclined plane.
- **Advisor:** [Hamda Ben Hadid](#) (Professor)
- **Co-supervisors:** [S  verine Millet](#) (Associate Professor), [Simon Dagois-Bohy](#) (Associate Professor)
- **Research Unit:** Laboratory of Fluid Mechanics and Acoustics (LMFA, UMR CNRS 5509)

Lyon 1 University Claude Bernard *September 2020 – September 2022*
Master's degree in Mechanics, Fluid Mechanics and Energetics track

Le Mans Universit   *September 2017 – June 2020*
Bachelor's degree in Engineering Sciences, Acoustics track

Research Activity

My research focuses on the stability of thin-film flows of non-Newtonian fluids down an inclined plane, relevant to numerous natural and industrial contexts (avalanches, mudflows, coating processes, etc.), and prone to becoming unstable. Given the diversity of rheological behaviours, my work aims to establish a unifying theoretical framework — independent of any specific rheological law — confronted with experimental data obtained on an inclined channel.



From left to right: surface waves on Carbopol, a surface coating process (Kistler & Schweizer (1997)), surface waves on melted chocolate.

International Conference Participation

11th International Bifurcation and Instabilities in Fluid Dynamics Symposium (BIFD)

July 6-9, 2026

L'Aquila, Italy

- **Presentation type:** Oral presentation
- **Title:** Short-wave Linear Stability of Non-Newtonian Fluids Down a Slope
- **Co-authors:** Fabien Depoilly, Séverine Millet, Simon Dagois-Bohy, Hamda Ben Hadid

GDR TransInter Workshop

June 10-12, 2025

Aussois, France

- **Presentation type:** Oral presentation
- **Title:** Stability of non newtonian fluids down a slope
- **Co-authors:** Fabien Depoilly, Simon Dagois-Bohy, Séverine Millet, François Rousset, Hamda Ben Hadid

10th International Bifurcation and Instabilities in Fluid Dynamics Symposium (BIFD)

June 24-28, 2024

Edinburgh, Scotland

- **Presentation type:** Oral presentation
- **Title:** Primary instability of roll waves on thin films of non-Newtonian fluids down a slope.
- **Co-authors:** Fabien Depoilly, Simon Dagois-Bohy, Séverine Millet, François Rousset, Hamda Ben Hadid

Fluids and Complexity III

December 6-8, 2023

Nice, France

- **Presentation type:** Poster presentation
- **Title:** Linear stability of a generalised Newtonian fluid down an incline: unifying the roll waves
- **Co-authors:** Fabien Depoilly, François Rousset, Séverine Millet, Hamda Ben Hadid, Simon Dagois-Bohy

Teaching Activities

- **Lyon 1 University Claude Bernard, Department of Mechanics**

- 2025: Introduction to Fluid Mechanics, 2nd-year Bachelor's in Mechanics (Tutorials)
- 2023 – 2025: Scientific Computing, 3rd-year Bachelor's in Mechanics (Labs)
- 2023 – 2025: Finite Difference Methods for PDEs in Mechanics, 1st-year Master's in Mechanics (Tutorials/Labs)

- **Polytech Lyon**

- 2025: Fluid Mechanics 2, 3rd-year Mechanical Engineering (Tutorials/Labs)
- 2025: Mathematical Tools for Engineers 2, 3rd-year Mechanical Engineering (Tutorials)
- 2023 – 2024: Basic Numerical Methods, 3rd-year Mechanical Engineering (Labs)
- 2023: Numerical Methods for Mechanics 1, 3rd-year Mechanical Engineering (Labs)

Responsibilities

Elected member of the Research Commission, Academic Council

2022 - 2024

Lyon 1 University Claude Bernard

Publications

- [1] Depoilly, F., Millet, S., Hadid, H. B., Dagois-Bohy, S., Rousset, F., “Unifying the roll waves,” *PLOS ONE*, vol. 19, no. 11, Nov. 19, 2024, Publisher: Public Library of Science, ISSN: 1932-6203. DOI: [10.1371/journal.pone.0310805](https://doi.org/10.1371/journal.pone.0310805) 